

ROUTER DEMO - Application Code

Step 1: Install React Router

```
npm install react-router-dom
```

Folder Structure

```
src/  
├── App.js  
├── components/  
│   ├── Navbar.js  
│   ├── ProtectedRoute.js  
├── pages/  
│   ├── Home.js  
│   ├── About.js  
│   ├── Contact.js  
│   ├── Dashboard.js  
│   ├── Profile.js  
│   ├── Settings.js  
│   ├── User.js  
│   ├── Login.js  
│   └── NotFound.js
```

App.js (Main Routing Setup)

```
import React from "react";  
import { BrowserRouter, Routes, Route } from "react-router-dom";  
  
import Navbar from "../components/Navbar";  
import ProtectedRoute from "../components/ProtectedRoute";  
  
import Home from "../pages/Home";  
import About from "../pages/About";  
import Contact from "../pages/Contact";  
import Dashboard from "../pages/Dashboard";  
import Profile from "../pages/Profile";  
import Settings from "../pages/Settings";  
import User from "../pages/User";  
import Login from "../pages/Login";
```

```

import NotFound from "../pages/NotFound";

function App() {
  return (
    <BrowserRouter>
      <Navbar />

      <Routes>
        {/* Basic Routes */}
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="/contact" element={<Contact />} />

        {/* Dynamic Route */}
        <Route path="/user/:id" element={<User />} />

        {/* Protected Route */}
        <Route path="/dashboard" element={
          <ProtectedRoute>
            <Dashboard />
          </ProtectedRoute>
        } />
        {/* Nested Routes */}
        <Route path="profile" element={<Profile />} />
        <Route path="settings" element={<Settings />} />
      </Route>

      {/* Login */}
      <Route path="/login" element={<Login />} />

      {/* 404 */}
      <Route path="*" element={<NotFound />} />
    </Routes>
  </BrowserRouter>
);
}

export default App;

```

```

import React from "react";
import { Link } from "react-router-dom";

function Navbar() {
  return (
    <nav>
      <Link to="/">Home</Link> |
      <Link to="/about">About</Link> |
      <Link to="/contact">Contact</Link> |
      <Link to="/dashboard">Dashboard</Link> |
      <Link to="/user/101">User 101</Link>
    </nav>
  );
}

export default Navbar;

```

ProtectedRoute.js

```

import React from "react";
import { Navigate } from "react-router-dom";

function ProtectedRoute({ children }) {
  const isLoggedIn = false; // change to true to test

  return isLoggedIn ? children : <Navigate to="/login" />;
}

export default ProtectedRoute;

```

Dashboard.js (Nested Routing)

```

import React from "react";
import { Link, Outlet } from "react-router-dom";

function Dashboard() {
  return (
    <div>
      <h2>Dashboard</h2>

```

```
    <Link to="profile">Profile</Link> |  
    <Link to="settings">Settings</Link>  
  
    <Outlet />  
  </div>  
);  
}  
  
export default Dashboard;
```

User.js (Dynamic Route)

```
import React from "react";  
import { useParams } from "react-router-dom";  
  
function User() {  
  const { id } = useParams();  
  
  return <h2>User ID: {id}</h2>;  
}  
  
export default User;
```

Pages

Home.js

```
export default function Home() {  
  return <h2>Home Page</h2>;  
}
```

About.js

```
export default function About() {  
  return <h2>About Page</h2>;  
}
```

Contact.js

```
export default function Contact() {  
  return <h2>Contact Page</h2>;  
}
```

Profile.js

```
export default function Profile() {  
  return <h3>Profile Page</h3>;  
}
```

Settings.js

```
export default function Settings() {  
  return <h3>Settings Page</h3>;  
}
```

Login.js

```
export default function Login() {  
  return <h2>Please Login</h2>;  
}
```

NotFound.js

```
export default function NotFound() {  
  return <h2>404 - Page Not Found</h2>;  
}
```